INCH-POUND

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5 September 2005
SUPERSEDING
A-A-58047B
7 December 2004

COMMERCIAL ITEM DESCRIPTION
SIGNAL GENERATOR
10 MHz to 26.5 GHz

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a 10 MHz to 26.5 GHz Signal Generator with AM, FM, pulse modulation (PM) and sweep capabilities. This CID is meant as a minimum requirement for the signal generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and bid sample testing, if so requested, within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 3 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Order of Precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this CID shall take precedence. Nothing in this CID; however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

2.3 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.3.1 Operating Temperature. The equipment shall conform to its specified performance and accuracy while being operated as a volt meter in a temperature range of 0° C to 50° C.

2.3.2 Non-Operating Temperature. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -40° C to 70° C.

Comments, suggestions, or questions on this document should be addressed to: 542 ATSG/GBTRND 380 Richard Ray Blvd, Ste 104, Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

FSC 6625

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2.4 Electrical Power Sources. The equipment shall operate from nominal single-phase commercial, military, and shipboard power sources of 110/220 volts AC (±10%) at line frequencies of 48 Hz to 63 Hz.. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein.

2.5 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 4,000 hours of operation with a statistical certainty of 95 %.

2.6 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours.

2.7 Performance Requirements. The instrument shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions, and operating service condition as specified in MIL-PRF-28800.

2.7.1 Frequency Range. The minimum frequency range of the signal generator shall be 10 MHz to 26.5 GHz.

2.7.1.1 Resolution. The generator shall have a frequency resolution of $\leq$ 0.1 Hz.

2.7.1.2 Accuracy. The generator accuracy shall be at least $\pm 1 \times 10^{-7}$ per year

2.7.1.3 Aging Rate. The generator aging rate shall be based on the internal reference oscillator aging rate which shall be at least $\pm 5 \times 10^{-9}$ per day or $\pm 1 \times 10^{-7}$ per year

2.7.2 Spectral Purity.

2.7.2.1 Harmonics. The generator harmonics shall be $\leq$ -35dBc over the entire frequency range

2.7.2.2 Sub-Harmonics. The sub-harmonics shall be as follows:

10 MHz to 18 GHz: $\leq$ -25 dBc
18 GHz to 26.5 GHz: $\leq$ -20 dBc

2.7.2.3 Spurious Non-Harmonics. The spurious non-harmonics shall be as follows:

10 MHz to 2 GHz: $\leq$ -40 dBc
2 GHz to 26.5 GHz: $\leq$ -50 dBc

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2.7.2.4 SSB Phase Noise. The generator shall have the following single sideband phase noise characteristics, at a 100kHz offset from the carrier:

10 MHz to 2 GHz: ≤ –100 dBc/Hz
2 GHz to 10 GHz : ≤ –104 dBc/Hz
10 GHz to 26.5 GHz : ≤ –97 dBc/Hz

2.7.3 RF Output. The minimum output level range into 50 ohm nominal shall be as follows:

10 MHz to 18 GHz: +8 dBm to -99 dBm
18 GHz to 20 GHz: +5 dBm to -99 dBm
20 GHz to 26.5 GHz: +1 dBm to -99 dBm

2.7.3.1 Output Resolution. The generator shall have an output resolution of ≤ 0.1 dB.

2.7.3.2 Output Protection. The generator shall have a reverse power protection to +20 dBm.

2.7.3.3 Output Connector. The RF output connector shall be APC-3.5 female or equivalent.

2.7.3.4 Output Accuracy. The output accuracy shall be ± 2dB.

2.7.4 Amplitude Modulation (AM).

2.7.4.1 AM Depth. The generator's amplitude modulation depth shall be at least 0 to 90%.

2.7.4.2 AM Accuracy. At a 1 kHz rate and 30% depth, the accuracy shall be ≤ (±6% of the setting +1%).

2.7.4.3 AM Bandwidth. At 30% depth with pulse off, the 3dB bandwidth shall be 10 kHz or better.

2.7.4.4 AM Distortion. At a 1 kHz rate and 50 % depth, the distortion shall be ≤ 5%.

2.7.5 Frequency Modulation (FM).

2.7.5.1 FM Rate. The generator shall have a frequency modulation rate to include 50 kHz to 1 MHz.

2.7.5.2 FM Deviation. The generator's frequency modulation maximum peak deviation shall include or exceed the following range:

500 MHz to 1 GHz: 0 to 2 MHz
1 GHz to 2 GHz: 0 to 4 MHz
2 GHz to 3.2 GHz: 0 to 8 MHz
≥ 3.2 GHz: 0 to 10MHz

2.7.5.3 FM Accuracy. The generator shall have a deviation accuracy of ≤ (±5% of deviation + 20 Hz).

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2.7.5.4 FM Sensitivity. The generator shall have a frequency modulation sensitivity of $\leq \pm 1 V_{peak}$.

2.7.6 Pulse Modulation (PM).

2.7.6.1 On/Off Ratio. The on/off ratio for the pulse modulation shall be 80 dB or greater.

2.7.6.2 Rise and Fall Time. The rise and fall time between 10% and 90% of the pulse shall be ≤ 25 nanoseconds above ≥ 125 MHz.

2.7.6.3 Repetition Rate. The generator shall provide pulse modulation with a repetition rate of ≥ 100 Hz to ≤ 1 MHz with a leveled output at frequencies ≥ 500 MHz.

2.7.6.4 Pulse Width. The minimum pulse width shall be at least 100 nanoseconds.

2.7.7 Sweep Mode. The generator shall have a sweep mode, analog and/or digital, that has a step sweep of frequency or amplitude, or both, from start to stop.

2.7.7.1 Sweep Time. The minimum sweep time shall be at least 45 milliseconds.

2.7.7.2 Sweep Range. The sweep shall include a selectable sweep range between any two power levels at a single continuous wave (CW) frequency.

2.7.7.3 Switching Speed. The generator shall the following switching speed:

≤ 20 GHz: ≤ 50 milliseconds
> 20 GHz: ≤ 70 milliseconds

2.7.7.4 Triggering. The generator shall have the ability to trigger externally, automatically, in a single mode, or as a result of a GPIB command.

2.7.7.5 Markers. The signal generator shall have at least 5 independent markers.

2.7.8 Remote interface. The generator shall come with an IEEE 488 standard interface.

2.8 Physical Parameters. The generator's component dimensions and weight shall be consistent with current commercial offerings. The equipment shall have the following physical characteristics.

2.8.1 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with four standard height units, 4U. One unit or "U" is defined by ANSI/EIA-310-D to be 1.75 inches or 44.45 mm.

2.8.2 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed a two-person lift of 74 pounds (33.6 kilograms).

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2.8.3 Rack-Mount. The analyzer shall be capable of being rack-mounted in a standard ANSI/EIA-310-D equipment rack.

2.9 Manuals. The equipment shall be delivered with operator, maintenance/service, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

If evidence substantiating that the Mean Time Between Failures (MTBF) of the instrument exceeds 20,000 hours is submitted, the maintenance/service manual shall not be required.

3. REGULATORY REQUIREMENTS.

3.1 Regulatory Requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 Acceptance. The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING.

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://assist.daps.dla.mil or from the Standardization Documents Order Desk, 700 Robbins Ave., Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.1.1 MIL-PRF-28800F, 24 June 1996, Test Equipment for Use with Electrical and Electronic Equipment, General Specification

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6.1.2 The Code of Federal Regulations, (CFR), may be obtained online at http://www.gpoaccess.gov/cfr/index.html or from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 Institute of Electrical and Electronics Engineers (IEEE) standards products can be ordered online at http://www.ieee.org or the IEEE Operations Center. (Phone: 1-800-678-IEEE). Mail: IEEE Operations Center, Sales Office, 445 Hoes Lane, PO Box 1331, Piscataway, NJ, 08855-1331, USA.

6.1.3.1 IEEE Std 488.2-1992, IEEE Standard Codes, Formats, Protocols, and Common Commands for Use With IEEE Std 488.1-1987, IEEE Standard Digital Interface for Programmable Instrumentation.

6.1.4 Electronics Industry Alliance (ANSI/EIA). Hardcopy of EIA standards and technical publications are available online at http://global.ihs.com or from the Global Engineering Documents, the exclusive distributor of EIA standards. Phone number: 1-800-854-7179.

6.1.4.1 EIA-310-D, Cabinets, Racks, Panels, and Associated Equipment.

6.2 Ordering data. Procurement documents must specify the title, number, and date of this CID, the issue of the ASSIST Online, http://assist.daps.dla.mil to be used in the solicitation, and preservation and packaging requirements.

6.3 Key words.

Calibration
Generator
Signal

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MILITARY INTEREST

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 84

Agent:
Air Force - 99

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